

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement 2.5V Diode SPICE Model Document

-Monte Carlo-

**Version 0.5 Phase 1
2016/12/22**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.5_p1	2016/12/22	Macro Chang	1. Original release. 2. This Monte Carlo model includes process variation characteristics extracted from model cards as below: l110ae_diode25_v051.mdl l110ae_diode25_v051.lib

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below, and the simulation results from other versions of the simulators might be different

Synopsys HSPICE release 2014.9SP2
Cadence SPECTRE version 15.1.0.257 64bit
Mentor Graphics ELDO version 15.4

1.3.1 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. The user can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo diode model
TT: Typical diode model
FF: Fast diode model
SS: Slow diode model

1.3.2 Sigma number for Monte-Carlo model

The user can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

Follow these steps to invoke the model file during simulation.

Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;

Example: `.param sigma=3`

Define the flag of Monte-Carlo simulation;

PROCESS: 1 to turn on the MC process simulation

0 to turn off the MC process simulation

Example: `.param process=1`

`.param process=0`

Define the number of Monte Carlo iterations and control MC simulations:

Example: `sweep monte=1000` (for HSPICE & ELDO)

`montecarlo numruns=1000` (for SPECTRE)

For Synopsys HSPICE

Copy following files into your simulation directory:

ModelFileName_mc.mdl

ModelFileName_mc_statistical_p.mdl

ModelFileName_mc.lib

ModelFileName_mc_corner.lib

- Add the following statement in the input file to the simulator for 90% dimension Shrinkage.

`.Option scale=0.9`

`.lib "/ModelFileName_mc_corner.lib" CaseName`

, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Xxx' statement for sub-circuit format model, or

'Dxx' statement for non sub-circuit format model

Example1:(for sub-circuit format model)

`X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5`

Example2:(for non sub-circuit format model)

`D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5`

For Cadence SPECTRE

Copy following files into your simulation directory:

ModelFileName_mc.mdl.scs
ModelFileName_mc_statistical_p.mdl.scs
ModelFileName_mc.lib.scs
ModelFileName_mc_corner.lib.scs

- Add the following statement in the input file to the simulator for 90% dimension Shrinkage.

SetOption1 options scale=0.9
include "./ModelFileName_mc_corner.lib.scs" section=CaseName
, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model

Example1:(for sub-circuit format model)
x1 (Va Vc) DeviceName area=1e-10 pj=4e-5
Example2:(for non sub-circuit format model)
d1 Va Vc DeviceName area=1e-10 pj=4e-5

For Mentor Graphics ELDO

Copy following files into your simulation directory:

ModelFileName_mc.mdl.eldo
ModelFileName_mc_statistical_p.mdl.eldo
ModelFileName_mc.lib.eldo
ModelFileName_mc_corner.lib.eldo

- Add the following statement in the input file to the simulator for 90% dimension Shrinkage.

.Option scale=0.9
.lib "./ModelFileName_mc_corner.lib.eldo" CaseName
, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model

Example1:(for sub-circuit format model)
X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5
Example2:(for non sub-circuit format model)
D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

ModelFileName: l110ae_diode25_v051
DeviceName: dion_25_l110ae
 diop_25_l110ae

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range(Before 90% shrinkage):

2.5V DION

0.5um<=WDES<=14.3um

5um<=LDES<=25um

2.5V DIOP

0.5um<=WDES<=14.3um

5um<=LDES<=25um

Temperature range:

-50 °C ~ +125 °C

2 Monte Carlo Simulation Results for Process Variation

Monte Carlo iterations is 1000 times to be performed in following simulation condition and the following corner points show TT, FF, and SS value, individually. Reasonable Monte Carlo iterative times are 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)^(-1/2).

2.1 Simulation result plots

The following figures show out DIODE device's process variation at 25 C. Area/Pj dimension is before shrinkage.

2.1.1 2.5V DION simulation results:

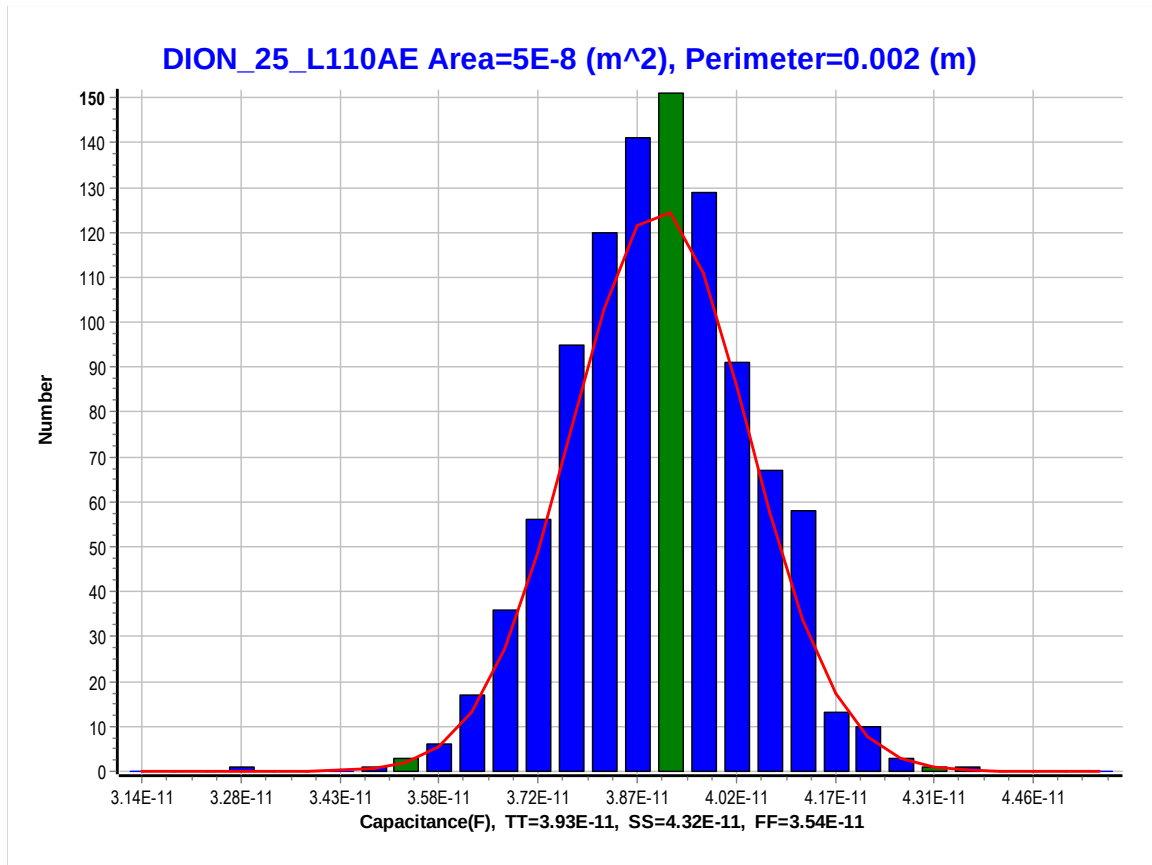


Figure 2-1 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"

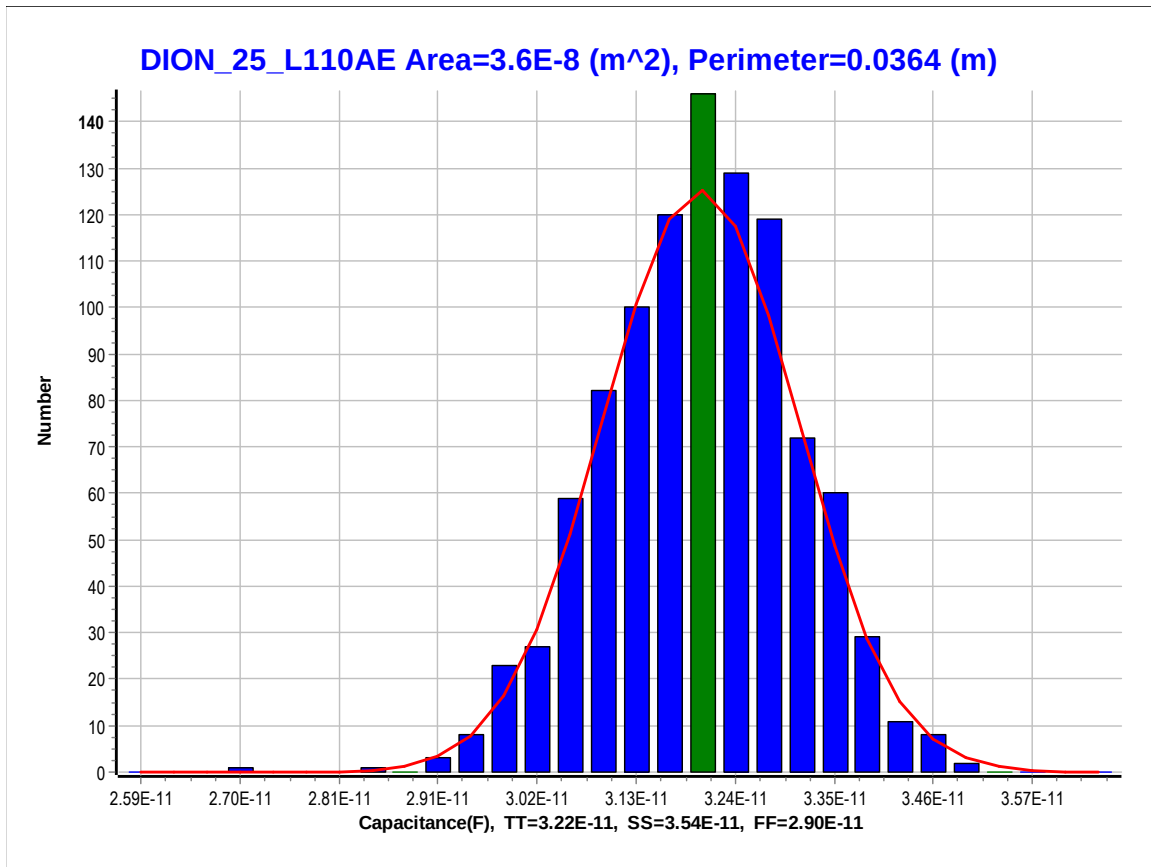


Figure 2-2 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V"

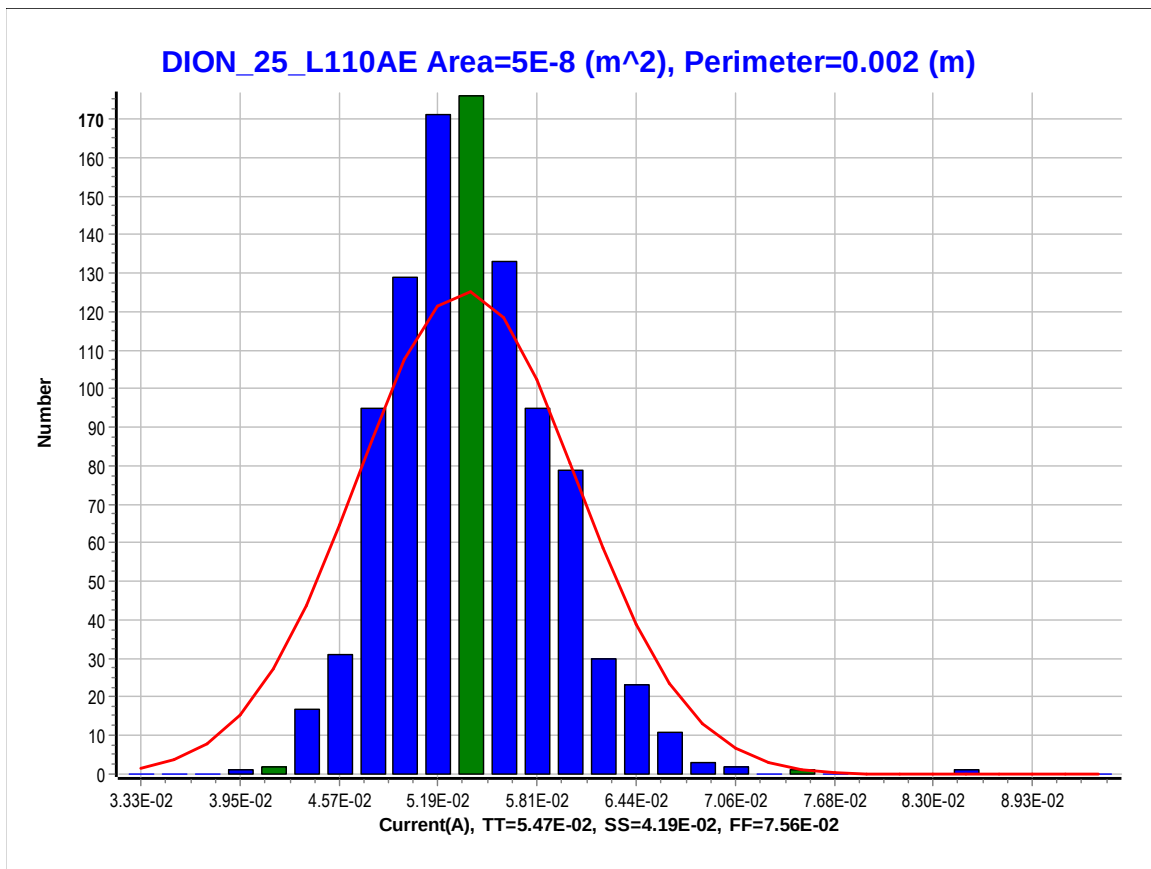


Figure 2-3 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"

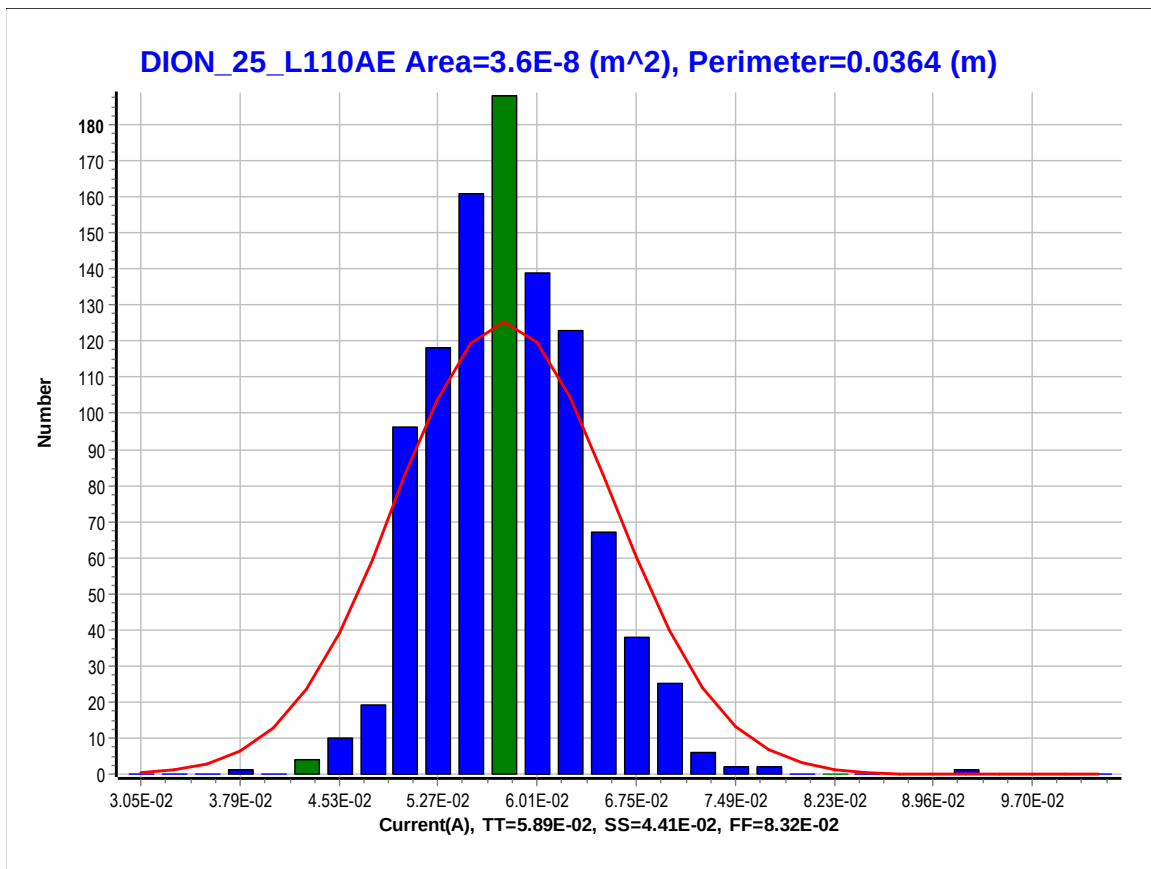


Figure 2-4 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"

2.1.2 2.5V DIOP simulation results:

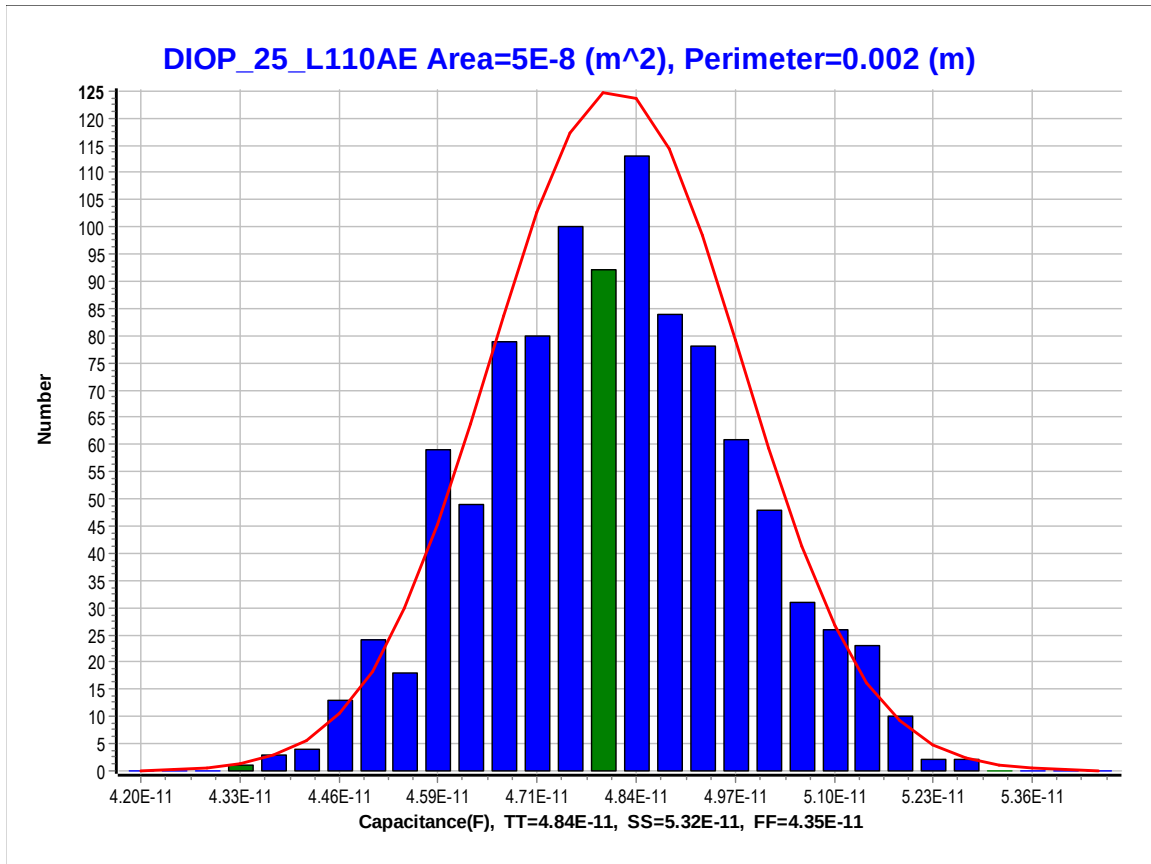


Figure 2-5 Device 1: Monte Carlo graph of "Capacitance @ Vbias=0V"

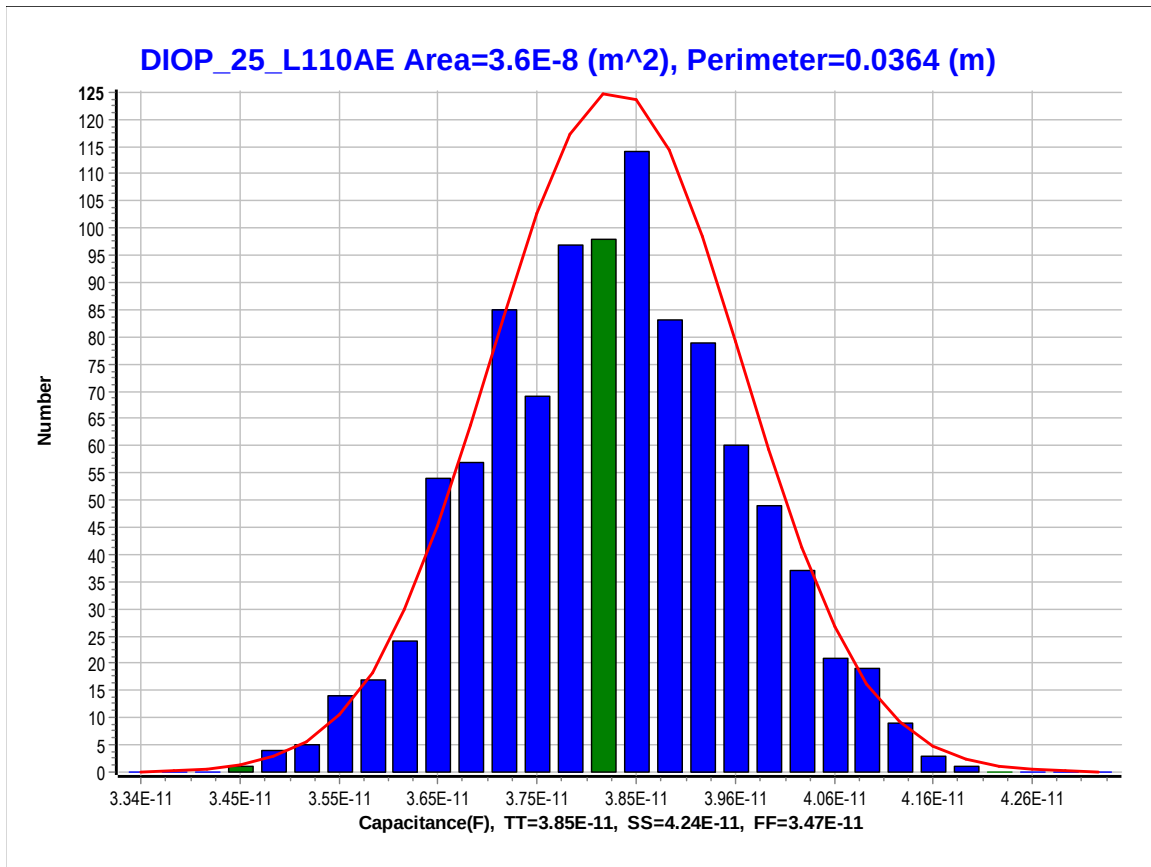


Figure 2-6 Device 2: Monte Carlo graph of "Capacitance @ Vbias=0V"

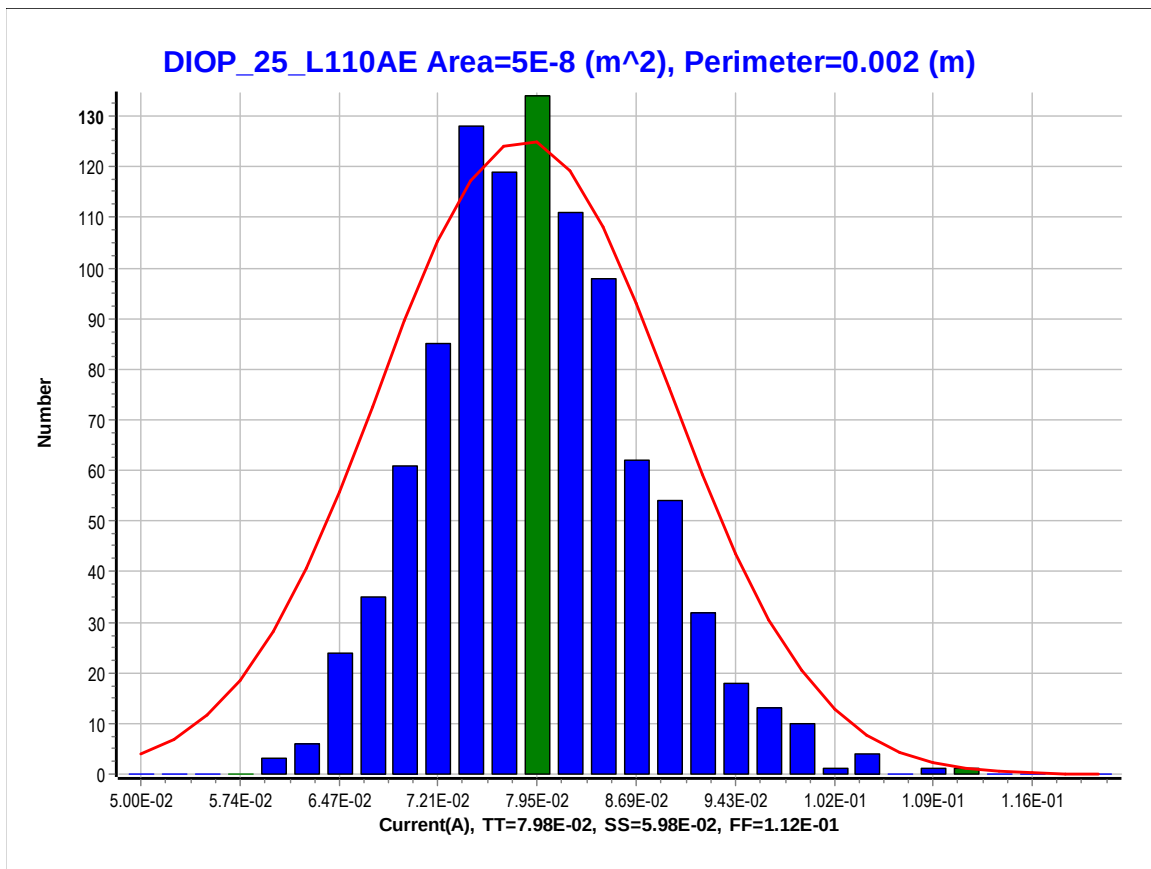


Figure 2-7 Device 1: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"

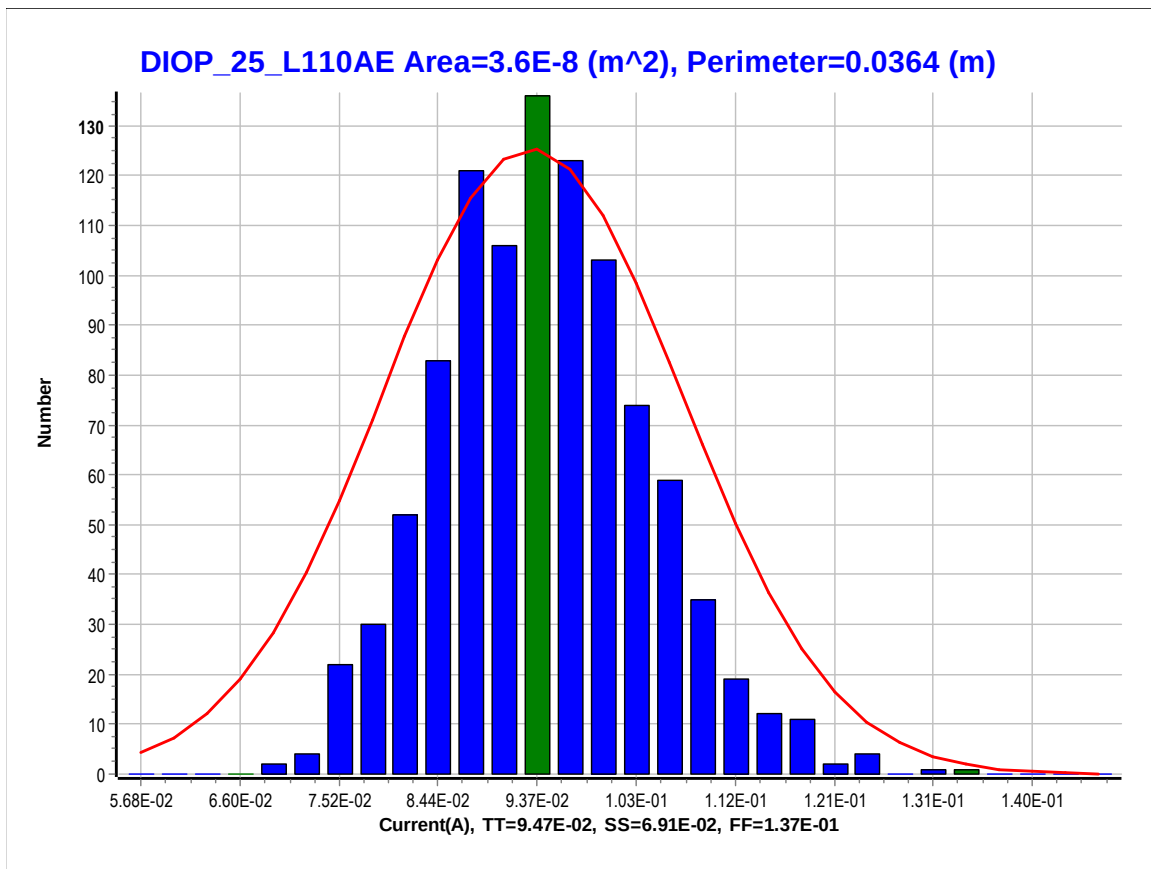


Figure 2-8 Device 2: Monte Carlo graph of "Forward Current @ Vbias= 1.0V"

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	None	None

3.2 ELDO Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	None	None

3.3 SPECTRE Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1.	None	None